

Supporting User Engagement in Testing, Auditing, and Contesting AI

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ABSTRACT

In recent years, interest in directly involving end users in testing, auditing, and contesting AI systems has grown. The involvement of end users, especially from diverse backgrounds, can be essential to overcome AI developers' blind spots and to surface issues that would otherwise go undetected prior to causing real-world harm. Emerging bodies of work in CSCW have begun to explore ways to engage end-users in testing and auditing AI systems, and to empower users to contest problematic or erroneous AI outputs. As this is a nascent area of work, we still know little about how to support effective user engagement. In this one-day workshop, we will bring together researchers and practitioners from academia, industry, and non-profit organizations to share ongoing efforts related to this workshop's theme. Central to our discussions will be the challenges encountered in developing tools and processes to support user involvement, strategies to incentivize involvement, the asymmetric power dynamic between AI developers and end users, and the role of regulation in enhancing the accountability of AI developers and ameliorating potential burdens towards end-users. Overall, we hope the workshop will help shape the future of user engagement in building more responsible AI.

KEYWORDS

algorithm auditing, human-centered AI, responsible AI, usability testing

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1 INTRODUCTION

Recent years have seen growing interest among both researchers and practitioners in directly engaging end users in testing or even systematically auditing algorithmic systems for errors or problematic behaviors [2, 5, 7, 9, 12, 15, 17, 19, 23–25, 28, 30, 32, 35]. The involvement of end users from diverse backgrounds can play a crucial role in overcoming AI developers' blind spots, and helping to surface issues that would otherwise go undetected before causing real-world harm [3, 5, 15, 17, 23, 24].

To this end, a growing body of research in CSCW and HCI has explored potential tools, processes, and guidelines that aim at engaging end-users in testing and auditing AI applications, and contesting AI outputs in domains like content moderation, image captioning and classification, and recommendation systems [5, 15–18, 23, 24, 31]. In parallel with these research efforts, several major technology companies have begun to experiment with approaches that engage users in testing and auditing their AI products and services for problematic behaviors [7, 25, 32, 35]. In 2021, Twitter introduced its first “algorithmic bias bounty” challenge to engage users in identifying harmful biases in its image cropping algorithm [7]. Inspired by “algorithmic bias bounty,” OpenAI initiated a series of feedback contests and red teaming efforts to engage end users in providing feedback on problematic model outputs during their interactions with the ChatGPT chatbot [1]. More recently, AI Village, a community of hackers and data scientists concerned with AI security, announced at the DEF CON hacker convention that it will hold the largest-ever public Generative AI red teaming exercise to engage users in testing and auditing generative AI models from a wide range of technology companies [32].

As this new body of work grows, such efforts to connect knowledgeable stakeholders is hugely valuable. In this workshop, we plan to bring together researchers and AI practitioners from academia, industry, and non-profit organizations to share their ongoing efforts around engaging end users in testing, auditing, and contesting AI systems. Through a series of talks, panels, presentations, and discussions, we will explore the following high-level questions:

- How can we better design and deploy tools, processes, and guidelines to engage end users in testing, auditing, and contesting AI systems?
- How can we support more effective communication and collaboration between AI developers, policy makers, and

end users throughout the process of design, development, and oversight of AI systems?

- What are some unique design considerations for user engagement at different scales of engagement (e.g., small-scale user testing vs. large-scale auditing) and different contexts (e.g., public interest AI deployed by public sector agencies vs. AI products and services developed by profit-driven companies)?
- What are risks are posed (to end users, AI system owners, and other stakeholders) when engaging users in testing, auditing, and contesting AI systems? How might we better identify and mitigate these risks?

2 WORKSHOP THEMES

Designing Tools and Processes to Support User Involvement

Prior work in CSCW and HCI has started to explore tools in supporting user engagement in AI testing, auditing, and contesting [2, 5, 16, 17, 28]. However, we still lack understanding on the design requirements and considerations to effectively engage users [9, 10, 19, 21, 34]. In addition, user involvement in testing and auditing AI systems often depends on custom or one-off systems that are at risk of becoming obsolete when platforms change and new technologies emerge [21, 26, 27, 31]. Finally, converting end-user feedback into AI model updates necessitates new tools and processes to foster collaboration between AI practitioners and end users [11, 34, 36]. In this workshop, we intend to discuss design patterns and methods for enhancing user involvement in AI testing, auditing, and contesting. Questions related to this theme includes:

- What are some unique design considerations for tools and processes to support user involvement in AI testing, auditing, and contesting?
- How might researchers develop testing and auditing strategies and systems that can serve communities in a long-lasting manner?
- How might researchers involve users throughout the AI design, development, and oversight lifecycle?
- How might researchers empower end users, especially those with low AI literacy [cf. 20], in contesting AI outputs?

Power Dynamics and Accountability

Recent research suggests a complex relationship between industry AI practitioners and end users, in the context of user engagement in AI testing and auditing. For users, organic efforts to test and audit AI systems often take on the form of activism [19, 21]. On one hand, emerging models of user-engagement in algorithm testing and auditing can empower users through the ability to directly connect with industry practitioners to try and identify issues together [9, 12, 17]. On the other hand, these models tend to firmly place the choice to take action with practitioners, potentially leaving users with less room for leverage via other means. Due to the inherent power dynamic between users and AI practitioners, users may worry that the issues they raise will not be adequately addressed. Indeed, companies may have profit incentives to prioritize addressing certain issues over others, independent of ethical considerations. Meanwhile, practitioners who open up their AI systems for early scrutiny may face challenges in fully trusting users not to

publicize issues before practitioners have had a chance to address them. Questions related to this theme includes:

- What are the potential strategies or tools to establish a more balanced power dynamic that fosters trust and collaboration between AI practitioners and end users?
- How might processes for user engagement be designed to give users real power, while fostering constructive dialogue with practitioners?
- What are some best practices for fostering a safe environment where end users and practitioners can work together in good faith, while maintaining transparency and accountability in the development and improvement of AI systems?

Promoting Synergy among Industry, Academia, and Public Regulation

In recent years, algorithm audits have risen to prominence as an approach to uncover biased, discriminatory, or otherwise harmful behaviors in algorithmic systems [4, 8, 14, 21, 27, 29, 33]. Although traditional academic expectations and industry practices have historically overlooked most end-user contributions (aside from sometimes crowdsourcing data from them) [6, 18, 29, 37], both industry practitioners and academic researchers have been experimenting with user-engaged algorithm testing and auditing [9]. Moreover, amidst the rapid developments and deployment of generative AI, current legislation and case laws, although aiming to empower users contesting AI outputs, often struggles to keep pace with technological innovation [13, 22]. To foster more responsible design, development, and oversight of these AI technologies, it is imperative to establish a collaborative framework that effectively combines resources and leverages the expertise of industry, academia, and public regulation. In this workshop, we will delve into the following questions to promote synergy across multiple sectors:

- How might CSCW researchers better support industry practitioners in facilitating user engagement in testing, auditing, and contesting AI products and services?
- What are some alternative approaches that CSCW researchers wish could become more prevalent in industry?
- What are some opportunities for regulation to guide and govern the current practices in user-engaged AI testing and auditing?
- What are some regulatory measures, public policies, or industry standards that could be implemented to achieve a more ethical prioritization that give users a greater voice in AI design, development, and oversight?

3 CALL FOR PARTICIPATION

We will recruit 40 - 45 participants (not including the organizers) who work on related areas in supporting user engagement in testing, auditing, and contesting AI. We will publicize the workshop through mailing lists, social media, and organizations that work on engaging users in AI testing, auditing, and contesting. Interested participants will be asked to contribute a brief statement of interest to the workshop. Submissions can take several forms, including: **1)** a 3-5 page position paper discussing or contributing to one or more themes highlighted in this proposal; **2)** 3-5 minute video or audio recording for the demo of an interactive system that is relevant to user-engagement in AI testing, auditing, and contesting; **3)** a case

study discussing ongoing work in engaging end users in AI testing, auditing, and contesting. Each submission will be reviewed in detail by 1-2 organizers and accepted based on quality of the submission and diversity of perspectives to allow for a meaningful exchange of knowledge between a broad range of stakeholders, upon review of the applicant pool by the committee as a whole.

4 WORKSHOP ACTIVITIES AND GOALS

The primary goal of this one-day, in-person workshop is to bring together researchers and AI practitioners from academia, industry, and non-profits to share their ongoing efforts around engaging end users in testing, auditing, and contesting AI systems. We will survey participants before the workshop to ask about their accessibility needs, workshop constraints, and their ideal takeaways from attending the workshop. We will use the results of this survey to guide the exact logistics and schedule of our workshop. However, we plan to organize the workshop around the following activities –

- **Welcome and Introduction (15 min):** Opening talk by one of the organizers, in which they will warmly welcome the participants, present the topic, and outline the format of the workshop session.
- **Author Presentation + Panel One (60 min):** Each participant will introduce themselves and their work. After the short presentation, each participant will join a panel facilitated by one of the organizers.
- **Coffee Break 30 min**
- **Author Presentation + Panel Two (60 min):** Each participant will introduce themselves and their work. After the short presentation, each participant will join a panel facilitated by one of the organizers.
- **Lunch Break (120 min)**
- **Design Workshop (group activity) (60 min):** Design workshop activities for 4-5 groups based on workshop themes each facilitated by the organizers. Note: we are open to updating the agenda and incorporating new topics based on the submissions from workshop participants and their backgrounds.
- **Coffee Break (15 min)**
- **Workshop Report (30 min):** Participants from each breakout group report back and discuss with the larger group.
- **End note (15 min):** We plan to create a Slack or Discord channel to foster ongoing communication and collaboration among participants, extending the discussion beyond the scope of this workshop session.

We will invite a keynote speaker and begin the workshop with a 30-minute keynote around their insights in user-engaged AI testing, auditing, and contesting. We will alternate between participant presentations, Q&A, smaller group activities and plenary discussions throughout the day. By facilitating interaction with different groups of attendees, participants will have ample opportunities to exchange ideas and perspectives. To streamline communication and coordination, we plan on using a website, a Slack channel, and a Google Drive shared with participants to coordinate activities before, during, and after the workshop. The website will serve as a central hub for distributing essential information, including the workshop description and proposal, pertinent related work, a call

for participation (complete with instructions and deadlines), the workshop agenda, and details about the keynote speaker and panelists. During the workshop, we will provide essential materials such as sticky notes, markers, and whiteboards to facilitate the design workshop activities. Furthermore, to ensure accessibility and ongoing documentation, a designated note-taker(s) will capture key points in a shared Google Doc. This approach will allow participants to review discussions and contributions even after the workshop concludes. Depending on the support from the conference organizers, we will include a variety of video-, audio-, and text-based ways of engaging to support participants with a diversity of visual, hearing, speech, and cognitive abilities.

5 ORGANIZERS

Wesley Hanwen Deng is a Ph.D. student at the Human-Computer Interaction Institute (HCII) at Carnegie Mellon University. Drawing on approaches from human-computer interaction, machine learning, design, and organizational science, his research focuses on supporting responsible AI practice in real-world contexts. His current works aim to scaffold effective communication and collaborations between AI developers and everyday users. He is currently supported by Jacob Foundation and The Digital Transformation and Innovation Center by PwC. Before joining CMU, Wesley completed his BA in Computer Science with the highest honor at UC Berkeley, where he also received an EECS research honor.

Michelle S. Lam is a Computer Science Ph.D. student at Stanford University in the Human-Computer Interaction Group. She builds systems that empower everyday users to actively reshape the design and evaluation of AI systems. She is supported by a Stanford Interdisciplinary Graduate Fellowship and has been a Stanford HAI Graduate Fellow, Stanford Technology and Racial Equity Graduate Fellow, Brown Institute for Media Innovation Magic Grantee, and Siebel Scholar. She holds an MS and BS in Computer Science from Stanford University.

Ángel Alexander Cabrera is a Ph.D. student in the Human-Computer Interaction Institute (HCII) at Carnegie Mellon University. He works on human-centered AI, specifically in applying techniques from HCI and visualization to help people better understand and improve their machine learning models. He is supported by an NSF Graduate Research Fellowship and have spent time at Apple AI/ML, Microsoft Research, and Google.

Danaë Metaxa is an Assistant Professor at the University of Pennsylvania in the Computer and Information Science department, with a secondary appointment in the Annenberg School for Communication. In their research, Dr. Metaxa studies bias and representation in algorithmic systems and content, focusing on high-stakes social settings like politics and employment, and in particular on the experiences of marginalized people. Along with Andrew Head, they are the co-founder of the Penn HCI group. Prior to joining Penn, Dr. Metaxa was a postdoctoral scholar with the Stanford Center of Philanthropy and Civil Society's Program on Democracy and the Internet, after receiving their Ph.D. in Computer Science from Stanford University.

Motahhare Eslami is an Assistant Professor at the School of Computer Science, Human-Computer Interaction Institute (HCII),

and Institute for Software Research (ISR), at Carnegie Mellon University. She earned her Ph.D. in Computer Science at the University of Illinois at Urbana-Champaign. Motahhare's research goal is to investigate the existing accountability challenges in algorithmic systems and to empower the users of algorithmic systems, particularly those who belong to marginalized communities or those whose decisions impact marginalized communities, make transparent, fair, and informed decisions in interaction with algorithmic systems.

Kenneth Holstein is an Assistant Professor in Carnegie Mellon University's Human-Computer Interaction Institute where he directs the CoALA Lab. His work explores human-AI complementarity, as well as how the design, development, evaluation, and oversight processes for human-algorithm decision systems can be improved in practice.

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